

# 实验3 IO虚拟化 实验报告

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2025-2026-1

## 1 调研部分

### 1.1 MMIO 与 PIO

#### 1.1.1 MMIO

MMIO 全称 Memory-Mapped I/O，即内存映射输入输出，核心思想是硬件设备的缓冲区被映射到主机的物理内存地址空间中。操作系统会为这些特定的物理地址区域建立页表映射，使得当 CPU 访问输入输出设备的地址时，其请求不会被发送到内存设备中，而是被重定向到相应的设备上。

**MMIO 的优点：**

- 编程方便：可以使用所有内存操作指令，进行位操作非常灵活。
- 性能较高：无需特殊的 I/O 指令，可以利用 CPU 的缓存和流水线机制。
- 地址空间大：共享整个 64 位的物理内存地址空间。

**MMIO 的缺点：**

- 占用地址空间：MMIO 区域会占用部分物理内存地址，导致系统可用的连续物理内存减少。
- 安全性：需要内存管理单元的保护，防止用户进程随意访问设备内存。

#### 1.1.2 PIO

PIO 全称 Port-Mapped I/O，即端口映射输入输出，核心思想是 CPU 拥有一个与内存地址空间完全分离的、独立的 I/O 地址空间。这个空间通常很小，设备在这个 I/O 空间中占据若干端口。

**PIO 的优点：**

- 不占用内存空间：I/O 空间独立，不会减少可用的物理内存地址。
- 指令明确：代码中具有特定的输入输出设备指令，明确是在和设备进行 I/O 操作，语义清晰。
- 历史兼容性：与古老的 x86 系列设备保持兼容。

**PIO 的缺点：**

- 编程不便：只能使用特定的输入输出指令，功能有限，无法直接进行位操作。若要修改内容，必须先读取整个端口的值，修改内容后再写回。
- 地址空间小：在 x86 系统中 I/O 的地址空间只有 64KB，可能不兼容复杂的输入输出设备。
- 性能较差：输入输出指令通常无法被很好地流水线化，且访问路径可能比内存访问更长。

## 1.2 设备枚举与设备发现

### 1.2.1 设备枚举的概念

设备枚举是指系统固件或操作系统在启动过程中，主动探测、识别和配置系统中所有硬件设备的过程。这个过程的目标是找出总线上连接的设备、读取设备的身份信息、为设备分配系统资源、加载必要的驱动程序使设备进入可用状态。最终，系统会构建内部的硬件资源映射，为操作系统的启动和运行奠定基础。

### 1.2.2 UEFI 与设备发现的过程

结合 UEFI 的设备发现过程：

#### 1. 起点：总线枚举

枚举通常从根开始，逐级向下进行。UEFI 会加载核心总线驱动程序，枚举器会遍历总线号、设备号、功能号，形成外部设备信息的数据结构。具体过程为，枚举器根据每个可能的数据结构，发送读取厂商 ID 的请求，如果收到合法的、有效的响应，就意味着该位置存在设备，枚举器会继续读取其配置空间寄存器，以确定设备的类型和功能，并为每个发现的设备分配内存和 I/O 空间，以及配置中断路由。

#### 2. 加载 UEFI 驱动程序

当枚举器识别出设备后，UEFI 的会尝试找到兼容的 UEFI 驱动程序，通过驱动程序为设备安装协议，从而允许操作系统与输入输出设备通过协议交互。

#### 3. 枚举其他总线

一旦核心的 PCI/PCIe 总线被枚举，其上的总线控制器设备本身也会被枚举，并触发新一级总线的枚举。例如通过 USB 主机控制器枚举连接的 USB 设备，通过 SATA AHCI 控制器枚举插入的硬盘或光驱。

#### 4. 构建硬件信息表并移交操作系统

UEFI 除本身使用枚举结果外，还负责将系统的硬件抽象信息传递给操作系统。UEFI 固件包含一个 ACPI 驱动程序，它会在启动后期收集所有枚举到的设备信息、资源分配情况，并生成 ACPI 表描述系统的硬件布局。此外，UEFI 还会将已加载的驱动和协议列表等信息，通过 EFI 系统表传递给操作系统载入程序。当操作系统载入程序启动后，就可以通过 UEFI 启动服务调用协议。当操作系统内核完全接管后，它会解析 ACPI 表，并基于此信息重新进行自己的设备枚举和驱动加载，然后退出 UEFI 启动服务。

## 2 实验目的

1. 学习参考书目，调研 IO 虚拟化的基础概念，查询相关资料并思考问题。
2. 启动 L2 虚拟机，在 QEMU 中添加模拟的 I/O 设备，在客户机环境安装对应驱动，并使用测试程序访问虚拟设备，分析客户机的内核信息。

## 3 实验步骤

1. 在头歌平台上启动虚拟环境，如图1所示

2. 切换到virtlab用户并进入实验目录 ~\labs\io\_lab 下, 如图2所示
3. 执行下面的指令, 启动 L2 虚拟机, 如图3所示

```
sudo qemu-system-x86_64 -smp 1 -m 2G -enable-kvm \  
-drive file=./ubuntu18.qcow2,if=virtio \  
-drive file=seed.iso,if=virtio \  
-device edu,dma_mask=0xFFFFFFFF \  
-net nic \  
-net user,hostfwd=tcp::2222-:22 \  
-nographic
```

4. 使用用户名 ubuntu, 密码 Virtlabs 登入 L2 虚拟机, 如图4所示
5. 使用命令 `uname -a` 确认 L2 虚拟机环境, 如图5所示
6. 进入 L2 的实验目录 ~\Book\Chapter4 下, 使用 `make` 命令编译源代码, 如图6所示
7. 使用命令 `sudo insmod pci.ko` 插入内核模块, 使用 `lspci` 查看 PCI 设备, 如图7所示
8. 使用命令 `cat /proc/devices | grep edu` 查看主设备号, 如图8所示, 主设备号为 245
9. 使用命令 `sudo mknod /dev/edu c 245 0` 构造设备结点, 并验证设备创建, 如图9所示
10. 使用命令 `sudo dmesg -c` 清除历史内核消息, `sudo ./test` 运行测试代码, `sudo dmesg` 查看新的内核消息, 如图10所示
11. 关闭 L2 虚拟机, 并释放实验资源

## 4 实验分析

### 4.1 内核模块实现

`pci.c` 是由 QEMU 模拟的 PCI 设备驱动, 创建字符设备并实现了文件操作和中断处理程序。驱动中 `ioctl` 的实现对应了用户空间的几个命令。

在 DMA 传输中, 使用了 `DMA_IRQ` 标志, 表示在 DMA 传输完成时产生中断。

中断处理程序 `irq_handler` 会读取中断状态寄存器 `IO_IRQ_STATUS`, 根据状态值打印不同的信息 (DMA 读中断、DMA 写中断、阶乘计算中断), 并确认中断 (写 `IO_IRQ_ACK` 寄存器)。

#### 4.1.1 设备驱动核心

```
static struct pci_device_id pci_ids[] = {  
    { PCI_DEVICE(QEMU_VENDOR_ID, EDU_DEVICE_ID), },  
    { 0, }  
};  
  
static struct pci_driver pci_driver = {  
    .name      = "edu_pci",  
    .id_table  = pci_ids,  
    .probe     = pci_probe,  
    .remove    = pci_remove,  
};
```

为 QEMU 虚拟设备提供驱动支持, 以在设备发现过程中提供信息。

### 4.1.2 设备接口

```
static struct file_operations fops = {  
    .owner      = THIS_MODULE,  
    .llseek     = llseek,  
    .read       = edu_read,  
    .write      = edu_write,  
    .open       = edu_open,  
    .unlocked_ioctl = edu_ioctl,  
};
```

定义文件操作的回调函数，用于描述驱动程序如何处理设备的打开、读写、关闭等文件相关操作，为 QEMU 设备驱动提供标准的接口实现方式，此后用户可通过文件操作与 QEMU 设备交互。

### 4.1.3 PRINT\_EDUINFO\_CMD 命令

```
for (i = 0; i < 64u; ++i) {  
    pci_read_config_byte(pdev, i, &val);  
    pr_info("config %x %x\n", i, val);  
}  
for (i = 0; i < 0x98; i += 4) {  
    pr_info("io %x %x\n", i, ioread32((void*)(mmio + i)));  
}
```

该命令用于打印 PCI 配置空间的信息和设备内存映射的 I/O 空间的内容。

### 4.1.4 SEND\_INTERRUPT\_CMD 命令

```
iowrite32(0x12345678, mmio + IO_IRQ_RAISE);
```

该命令通过写 IO\_IRQ\_RAISE 寄存器触发中断，用于测试中断功能。

### 4.1.5 FACTORIAL\_CMD 命令

```
iowrite32(0x80, mmio + IO_FACTORIA_IRQ);  
iowrite32(0xA, mmio + FACTORIA_VAL);  
ioread32((void*)(mmio + FACTORIA_VAL));
```

该命令通过写 IO\_FACTORIA\_IRQ 寄存器触发中断，向 FACTORIA\_VAL 寄存器写入参数 10 后，虚拟设备计算阶乘，通过读取 FACTORIA\_VAL 寄存器获取结果。

### 4.1.6 DMA\_WRITE\_CMD 命令

```
iowrite32((uintptr_t)(mmio+0x100), mmio + IO_DMA_DST);  
iowrite32(DMA_BASE, mmio + IO_DMA_SRC);  
iowrite32(4, mmio + IO_DMA_CNT);  
iowrite32(DMA_CMD|DMA_FROM_MEM|DMA_IRQ, mmio + IO_DMA_CMD);
```

该命令向虚拟设备提供目标地址、源地址、传输大小后，启动 DMA 读操作，DMA\_FROM\_MEM 表示数据从内存到设备。

#### 4.1.7 DMA\_READ\_CMD 命令

```
iowrite32(DMA_BASE, mmio + IO_DMA_DST);  
iowrite32((uintptr_t)(mmio+0x104), mmio + IO_DMA_SRC);  
iowrite32(4, mmio + IO_DMA_CNT);  
iowrite32(DMA_CMD|DMA_TO_MEM|DMA_IRQ, mmio + IO_DMA_CMD);
```

该命令向虚拟设备提供目标地址、源地址、传输大小后，启动 DMA 写操作，DMA\_TO\_MEM 表示数据从内存到设备。

#### 4.1.8 中断处理程序

```
static irqreturn_t irq_handler(int irq, void *dev)  
{  
    u32 irq_status = ioread32(mmio + IO_IRQ_STATUS);  
    if(irq_status == 0x100) pr_info("receive a DMA read interrupter!\n");  
    else if(irq_status == 0x101) pr_info("receive a DMA write interrupter!\n");  
    else if(irq_status == 0x1) pr_info("receive a FACTORIAL interrupter!\n");  
    iowrite32(irq_status, mmio + IO_IRQ_ACK);  
    return IRQ_HANDLED;  
}
```

中断处理程序根据中断状态寄存器 IO\_IRQ\_STATUS 的值，输出中断类型信息，并确认中断。

### 4.2 用户空间程序

test.c 通过打开 /dev/edu 设备文件并执行一系列 ioctl 命令来测试驱动功能，包括：

- fd = open("/dev/edu", O\_RDWR, S\_IRUSR | S\_IWUSR);: 打开设备
- PRINT\_EDUINFO\_CMD 打印设备信息。
- SEND\_INTERRUPT\_CMD 触发一个中断，驱动会处理这个中断并打印中断状态。
- FACTORIAL\_CMD 计算阶乘，驱动会打印计算结果。
- DMA\_WRITE\_CMD 执行 DMA 写操作（从内存到设备），传输完成后会产生中断。
- close(fd);fd=open("/dev/edu",O\_RDWR,S\_IRUSR|S\_IWUSR); 关闭设备重新打开
- DMA\_READ\_CMD: 执行 DMA 读操作（从设备到内存），传输完成后会产生中断。

## 5 实验总结

在本实验中，我复现了 QEMU 虚拟输入输出设备的创建，并通过 PCI 设备驱动与 QEMU 虚拟设备交互，实现了设备信息查询、中断触发、DMA 传输和阶乘计算等功能，收获颇丰。感谢课程组精心设计的虚拟化实验，也感谢老师和助教一直以来的付出和答疑解惑。

## A 内核消息

内核消息依次展示了 QEMU 设备的配置信息、寄存器信息、中断处理等内容，如下所示。

```
[ 121.380186] length 100000
[ 121.380252] config 0 34
[ 121.380292] config 1 12
[ 121.380332] config 2 e8
[ 121.380371] config 3 11
[ 121.380410] config 4 3
[ 121.380450] config 5 1
[ 121.380489] config 6 10
[ 121.380528] config 7 0
[ 121.380567] config 8 10
[ 121.380605] config 9 0
[ 121.380645] config a ff
[ 121.380684] config b 0
[ 121.380723] config c 0
[ 121.380768] config d 0
[ 121.380808] config e 0
[ 121.380847] config f 0
[ 121.380885] config 10 0
[ 121.380924] config 11 0
[ 121.380963] config 12 a0
[ 121.381002] config 13 fe
[ 121.381042] config 14 0
[ 121.381079] config 15 0
[ 121.381119] config 16 0
[ 121.381158] config 17 0
[ 121.381197] config 18 0
[ 121.381236] config 19 0
[ 121.381275] config 1a 0
[ 121.381314] config 1b 0
[ 121.381352] config 1c 0
[ 121.381391] config 1d 0
[ 121.381431] config 1e 0
[ 121.381470] config 1f 0
[ 121.381509] config 20 0
[ 121.381548] config 21 0
[ 121.381588] config 22 0
[ 121.381627] config 23 0
[ 121.381665] config 24 0
[ 121.381703] config 25 0
[ 121.381746] config 26 0
[ 121.381784] config 27 0
[ 121.381823] config 28 0
[ 121.381868] config 29 0
[ 121.381907] config 2a 0
```



```
[ 121.381946] config 2b 0
[ 121.381985] config 2c f4
[ 121.382025] config 2d 1a
[ 121.382064] config 2e 0
[ 121.382103] config 2f 11
[ 121.382142] config 30 0
[ 121.382181] config 31 0
[ 121.382219] config 32 0
[ 121.382258] config 33 0
[ 121.382315] config 34 40
[ 121.382356] config 35 0
[ 121.382398] config 36 0
[ 121.382438] config 37 0
[ 121.382479] config 38 0
[ 121.382520] config 39 0
[ 121.382573] config 3a 0
[ 121.382614] config 3b 0
[ 121.382653] config 3c b
[ 121.382692] config 3d 1
[ 121.382731] config 3e 0
[ 121.382773] config 3f 0
[ 121.382774] dev->irq a
[ 121.382819] io 0 10000ed
[ 121.382840] io 4 0
[ 121.382862] io 8 0
[ 121.382884] io c ffffffff
[ 121.382906] io 10 ffffffff
[ 121.382928] io 14 ffffffff
[ 121.382949] io 18 ffffffff
[ 121.382971] io 1c ffffffff
[ 121.382993] io 20 0
[ 121.383018] io 24 0
[ 121.383038] io 28 ffffffff
[ 121.383060] io 2c ffffffff
[ 121.383081] io 30 ffffffff
[ 121.383103] io 34 ffffffff
[ 121.383125] io 38 ffffffff
[ 121.383146] io 3c ffffffff
[ 121.383168] io 40 ffffffff
[ 121.383189] io 44 ffffffff
[ 121.383211] io 48 ffffffff
[ 121.383233] io 4c ffffffff
[ 121.383254] io 50 ffffffff
[ 121.383276] io 54 ffffffff
[ 121.383298] io 58 ffffffff
[ 121.383319] io 5c ffffffff
[ 121.383341] io 60 ffffffff
```



```
[ 121.383361] io 64 ffffffff
[ 121.383383] io 68 ffffffff
[ 121.383407] io 6c ffffffff
[ 121.383429] io 70 ffffffff
[ 121.383451] io 74 ffffffff
[ 121.383473] io 78 ffffffff
[ 121.383494] io 7c ffffffff
[ 121.383516] io 80 0
[ 121.383537] io 84 ffffffff
[ 121.383559] io 88 0
[ 121.383581] io 8c ffffffff
[ 121.383603] io 90 0
[ 121.383624] io 94 ffffffff
[ 121.383677] irq_handler irq = 10 dev = 245 irq_status = 12345678
[ 122.400285] receive a FACTORIAL interrupter!
[ 122.400287] irq_handler irq = 10 dev = 245 irq_status = 1
[ 123.424372] computing result 375f00
[ 123.524728] receive a DMA read interrupter!
[ 123.524730] irq_handler irq = 10 dev = 245 irq_status = 100
[ 125.547930] receive a DMA read interrupter!
[ 125.547932] irq_handler irq = 10 dev = 245 irq_status = 100
```





```
* Support: https://ubuntu.com/advantage

System information as of Thu Nov 13 06:36:38 PM CST 2025

System load: 0.701171875      Processes: 114
Usage of /: 49.2% of 39.07GB  Users logged in: 0
Memory usage: 11%            IPv4 address for enp1s0: 10.0.2.2
Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Oct 7 23:12:08 2023 from 100.86.166.233
root@virtlab:~#
```

图 1 启动虚拟环境

```
System information as of Thu Nov 13 06:36:38 PM CST 2025

System load: 0.701171875      Processes: 114
Usage of /: 49.2% of 39.07GB  Users logged in: 0
Memory usage: 11%            IPv4 address for enp1s0: 10.0.2.2
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2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Oct 7 23:12:08 2023 from 100.86.166.233
root@virtlab:~# su - virtlab
virtlab@virtlab:~$ cd labs/io_lab
virtlab@virtlab:~/labs/io_lab$
```

图 2 进入实验目录



```
[ OK ] Reached target Login Prompts.
[ OK ] Started OpenBSD Secure Shell server.
      Starting Authorization Manager...
[ OK ] Started LSB: automatic crash report generation.
[ OK ] Started LSB: Record successful boot for GRUB.
[ OK ] Started Authorization Manager.
[ OK ] Started Accounts Service.
[ OK ] Started Dispatcher daemon for systemd-networkd.
[ OK ] Started LXD - container startup/shutdown.
[ OK ] Started Snap Daemon.
      Starting Wait until snapd is fully seeded...
      Starting Time & Date Service...
[ OK ] Started Time & Date Service.
[ OK ] Started Wait until snapd is fully seeded.
      Starting Apply the settings specified in cloud-config...
[ 21.946241] cloud-init[1036]: Cloud-init v. 23.1.2-0ubuntu0~18.04.1 running 'modules.
[ OK ] Started Apply the settings specified in cloud-config.
[ OK ] Reached target Multi-User System.
[ OK ] Reached target Graphical Interface.
      Starting Update UTMP about System Runlevel Changes...
      Starting Execute cloud user/final scripts...
[ OK ] Started Update UTMP about System Runlevel Changes.
[ 22.575405] cloud-init[1079]: Cloud-init v. 23.1.2-0ubuntu0~18.04.1 running 'modules.
[ 22.588496] cloud-init[1079]: Cloud-init v. 23.1.2-0ubuntu0~18.04.1 finished at Thu,
[ OK ] Started Execute cloud user/final scripts.
      Starting Daily apt download activities...
[ OK ] Reached target Cloud-init target.

Ubuntu 18.04.6 LTS ubuntu ttyS0

ubuntu login: █
```

图 3 启动 L2 虚拟机

```
ubuntu
Password: [ OK ] Started Daily apt upgrade and clean activities.

Last login: Sat Oct  7 06:07:02 UTC 2023 on ttyS0
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-213-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Thu Nov 13 10:47:11 UTC 2025

System load:  0.91           Processes:      83
Usage of /:   19.8% of 9.51GB Users logged in:  0
Memory usage: 7%            IP address for ens3: 10.0.2.15
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

29 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

[ OK ] Created slice User Slice of ubuntu.
      Starting User Manager for UID 1000...
[ OK ] Started Session 1 of user ubuntu.
[ OK ] Started User Manager for UID 1000.
ubuntu@ubuntu:~$ █
```

图 4 登录 L2 虚拟机



```
Last login: Sat Oct 7 06:07:02 UTC 2023 on ttyS0
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-213-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

System information as of Thu Nov 13 10:47:11 UTC 2025

System load:  0.91           Processes:      83
Usage of /:   19.8% of 9.51GB Users logged in:   0
Memory usage: 7%           IP address for ens3: 10.0.2.15
Swap usage:   0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

29 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

[ OK ] Created slice User Slice of ubuntu.
       Starting User Manager for UID 1000...
[ OK ] Started Session 1 of user ubuntu.
[ OK ] Started User Manager for UID 1000.
ubuntu@ubuntu:~$ [ OK ] Started Message of the Day.
uname -a
Linux ubuntu 4.15.0-213-generic #224-Ubuntu SMP Mon Jun 19 13:30:12 UTC 2023 x86_64 x86_64
ubuntu@ubuntu:~$
```

图 5 确认虚拟环境

```
extern void iowrite32(u32, void __iomem *);
~~~~~
/home/ubuntu/Book/Chapter4/pci.c:120:23: warning: passing argument 1 of 'iowrite32' makes
iowrite32(mmio+0x104, mmio + IO_DMA_SRC);
~~~~~
In file included from ./arch/x86/include/asm/io.h:223:0,
                 from ./arch/x86/include/asm/realmode.h:15,
                 from ./arch/x86/include/asm/acpi.h:33,
                 from ./arch/x86/include/asm/fixmap.h:29,
                 from ./arch/x86/include/asm/apic.h:10,
                 from ./arch/x86/include/asm/smp.h:13,
                 from ./arch/x86/include/asm/mmzone_64.h:11,
                 from ./arch/x86/include/asm/mmzone.h:5,
                 from ./include/linux/mmzone.h:912,
                 from ./include/linux/gfp.h:6,
                 from ./include/linux/idr.h:16,
                 from ./include/linux/kernfs.h:14,
                 from ./include/linux/sysfs.h:16,
                 from ./include/linux/kobject.h:21,
                 from ./include/linux/cdev.h:5,
                 from ./home/ubuntu/Book/Chapter4/pci.c:1:
./include/asm-generic/iomap.h:42:13: note: expected 'u32 {aka unsigned int}' but argument 1
extern void iowrite32(u32, void __iomem *);
~~~~~
Building modules, stage 2.
MODPOST 1 modules
CC      /home/ubuntu/Book/Chapter4/pci.mod.o
LD [M]  /home/ubuntu/Book/Chapter4/pci.ko
make[1]: Leaving directory '/usr/src/linux-headers-4.15.0-213-generic'
cc test.c -o test
ubuntu@ubuntu:~/Book/Chapter4$
```

图 6 编译源代码



```
CC      /home/ubuntu/Book/Chapter4/pci.mod.o
LD [M]  /home/ubuntu/Book/Chapter4/pci.ko
make[1]: Leaving directory '/usr/src/linux-headers-4.15.0-213-generic'
cc test.c -o test
ubuntu@ubuntu:~/Book/Chapter4$ sudo insmod pci.ko
ubuntu@ubuntu:~/Book/Chapter4$ lspci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.1 IDE interface: Intel Corporation 82371SB PIIX3 IDE [Natoma/Triton II]
00:01.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 03)
00:02.0 VGA compatible controller: Device 1234:1111 (rev 02)
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev)
00:04.0 Unclassified device [00ff]: Device 1234:11e8 (rev 10)
00:05.0 SCSI storage controller: Red Hat, Inc. Virtio block device
00:06.0 SCSI storage controller: Red Hat, Inc. Virtio block device
ubuntu@ubuntu:~/Book/Chapter4$ lspci -s 0:04.0 -vvv -xxxx
00:04.0 Unclassified device [00ff]: Device 1234:11e8 (rev 10)
    Subsystem: Red Hat, Inc. Device 1100
    Physical Slot: 4
    Control: I/O+ Mem+ BusMaster- SpecCycle- MemWINV- VGASnoop- ParErr- Stepping- S-
    Status: Cap+ 66MHz- UDF- FastB2B- ParErr- DEVSEL=fast >TAbort- <TAbort- <MAbort-
    Interrupt: pin A routed to IRQ 10
    Region 0: Memory at fea00000 (32-bit, non-prefetchable) [size=1M]
    Capabilities: <access denied>
    Kernel driver in use: edu_pci
00: 34 12 e8 11 03 01 10 00 10 00 ff 00 00 00 00 00
10: 00 00 a0 fe 00 00 00 00 00 00 00 00 00 00 00 00
20: 00 00 00 00 00 00 00 00 00 00 00 00 00 f4 1a 00 11
30: 00 00 00 00 40 00 00 00 00 00 00 00 00 0b 01 00 00
ubuntu@ubuntu:~/Book/Chapter4$
```

图 7 查看 PCI 设备

```
make[1]: Leaving directory '/usr/src/linux-headers-4.15.0-213-generic'
cc test.c -o test
ubuntu@ubuntu:~/Book/Chapter4$ sudo insmod pci.ko
ubuntu@ubuntu:~/Book/Chapter4$ lspci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.1 IDE interface: Intel Corporation 82371SB PIIX3 IDE [Natoma/Triton II]
00:01.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 03)
00:02.0 VGA compatible controller: Device 1234:1111 (rev 02)
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev)
00:04.0 Unclassified device [00ff]: Device 1234:11e8 (rev 10)
00:05.0 SCSI storage controller: Red Hat, Inc. Virtio block device
00:06.0 SCSI storage controller: Red Hat, Inc. Virtio block device
ubuntu@ubuntu:~/Book/Chapter4$ lspci -s 0:04.0 -vvv -xxxx
00:04.0 Unclassified device [00ff]: Device 1234:11e8 (rev 10)
    Subsystem: Red Hat, Inc. Device 1100
    Physical Slot: 4
    Control: I/O+ Mem+ BusMaster- SpecCycle- MemWINV- VGASnoop- ParErr- Stepping- S-
    Status: Cap+ 66MHz- UDF- FastB2B- ParErr- DEVSEL=fast >TAbort- <TAbort- <MAbort-
    Interrupt: pin A routed to IRQ 10
    Region 0: Memory at fea00000 (32-bit, non-prefetchable) [size=1M]
    Capabilities: <access denied>
    Kernel driver in use: edu_pci
00: 34 12 e8 11 03 01 10 00 10 00 ff 00 00 00 00 00
10: 00 00 a0 fe 00 00 00 00 00 00 00 00 00 00 00 00
20: 00 00 00 00 00 00 00 00 00 00 00 00 00 f4 1a 00 11
30: 00 00 00 00 40 00 00 00 00 00 00 00 00 0b 01 00 00
ubuntu@ubuntu:~/Book/Chapter4$ cat /proc/devices | grep edu
245 pci_edu
ubuntu@ubuntu:~/Book/Chapter4$
```

图 8 查看驱动信息



```
00:04.0 Unclassified device [00ff]: Device 1234:11e8 (rev 10)
Subsystem: Red Hat, Inc. Device 1100
Physical Slot: 4
Control: I/O+ Mem+ BusMaster- SpecCycle- MemWINV- VGASnoop- ParErr- Stepping- S-
Status: Cap+ 66MHz- UDF- FastB2B- ParErr- DEVSEL=fast >TAbort- <TAbort- <MAbort-
Interrupt: pin A routed to IRQ 10
Region 0: Memory at fea00000 (32-bit, non-prefetchable) [size=1M]
Capabilities: <access denied>
Kernel driver in use: edu_pci
00: 34 12 e8 11 03 01 10 00 10 00 ff 00 00 00 00 00
10: 00 00 a0 fe 00 00 00 00 00 00 00 00 00 00 00
20: 00 00 00 00 00 00 00 00 00 00 00 00 f4 1a 00 11
30: 00 00 00 00 40 00 00 00 00 00 00 00 0b 01 00 00

ubuntu@ubuntu:~/Book/Chapter4$ cat /proc/devices | grep edu
245 pci_educ
ubuntu@ubuntu:~/Book/Chapter4$ sudo mknod /dev/edu c <major> 0
-bash: major: No such file or directory
ubuntu@ubuntu:~/Book/Chapter4$ ls /dev/edu
ls: cannot access '/dev/edu': No such file or directory
ubuntu@ubuntu:~/Book/Chapter4$ sudo mknod /dev/edu c 245 pci_educ 0
mknod: extra operand `0'
Try 'mknod --help' for more information.
ubuntu@ubuntu:~/Book/Chapter4$ sudo mknod /dev/edu c 245 pci_educ
mknod: invalid minor device number a
ubuntu@ubuntu:~/Book/Chapter4$ ls /dev/edu
ls: cannot access '/dev/edu': No such file or directory
ubuntu@ubuntu:~/Book/Chapter4$ sudo mknod /dev/edu c 245 0
ubuntu@ubuntu:~/Book/Chapter4$ ls /dev/edu
/dev/edu
ubuntu@ubuntu:~/Book/Chapter4$
```

图 9 构造设备节点

```
[ 634.615390] io 40 ffffffff
[ 634.615400] io 44 ffffffff
[ 634.615410] io 48 ffffffff
[ 634.615420] io 4c ffffffff
[ 634.615430] io 50 ffffffff
[ 634.615440] io 54 ffffffff
[ 634.615450] io 58 ffffffff
[ 634.615459] io 5c ffffffff
[ 634.615469] io 60 ffffffff
[ 634.615479] io 64 ffffffff
[ 634.615489] io 68 ffffffff
[ 634.615499] io 6c ffffffff
[ 634.615509] io 70 ffffffff
[ 634.615519] io 74 ffffffff
[ 634.615529] io 78 ffffffff
[ 634.615539] io 7c ffffffff
[ 634.615549] io 80 0
[ 634.615559] io 84 ffffffff
[ 634.615569] io 88 0
[ 634.615578] io 8c ffffffff
[ 634.615588] io 90 0
[ 634.615599] io 94 ffffffff
[ 634.615634] irq_handler irq = 10 dev = 245 irq_status = 12345678
[ 635.648421] receive a FACTORIAL interrupter!
[ 635.648425] irq_handler irq = 10 dev = 245 irq_status = 1
[ 636.672318] computing result 375f00
[ 636.772674] receive a DMA read interrupter!
[ 636.772696] irq_handler irq = 10 dev = 245 irq_status = 100
[ 638.796718] receive a DMA read interrupter!
[ 638.796735] irq_handler irq = 10 dev = 245 irq_status = 100
ubuntu@ubuntu:~/Book/Chapter4$
```

图 10 查看内核消息